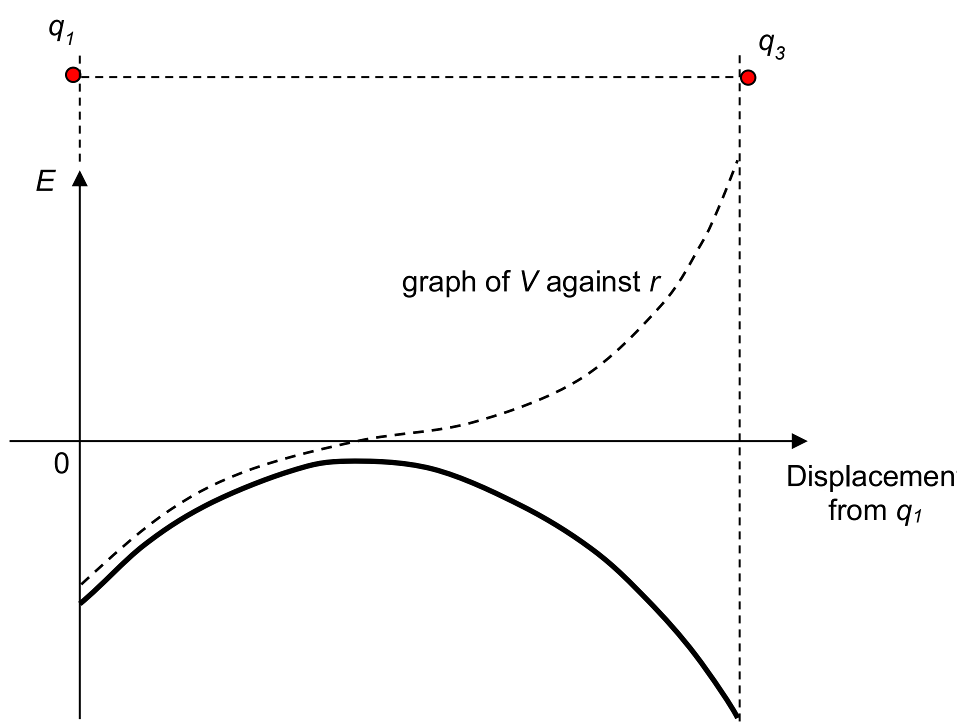
4	**(a)**	**(i)**	Electric potential is the work done **per unit positive charge** by an external force in bringing a test charge from infinity to that point without a change in kinetic energy.	2
		(ii)	Electric field strength at a point in the electric field is numerically equal to the potential gradient at that point, and directed towards lower potential. (no marks if direction not stated). or $E = -\frac{dV}{dr}$ (where E is the electric field strength at the point, V is the potential and r is the distance from a particular reference point)	1
	(b)		q_1, q_2 are negative. Potential around them is negative and increases (ie less negative) with distance from either charge. q_3 is positive. The potential around it is positive and decreases (ie less positive) with distance from the charge.	1 1
	(c)	**(i)**	Direction of field strength is perpendicular to P, radially towards q_2.	1
		(ii)	$\lvert E\rvert = \Delta V / \Delta r = [-1 - (-3)] / (1.5 \times 10^{-2}) = 130 \text{ V m}^{-1}$ (2 sf)	1
	(d)		Work done by external agent $= q\,(V_f - V_i) \quad = q\,(V_R - V_P)$ $= (-1.60 \times 10^{-19})[(+3.0) - (-2.0)]$ $= -8.0 \times 10^{-19}$ J	 1 1
	(e)		 • 1 mark for correct shape (having the turning point vs the inflexion point) • 1 mark for negative and not touching the x-axis • 1 mark for a larger magnitude of E at q_3 than at q_1	3

Comments

1. Students should first draw the potential V-curve, then use $E=-\frac{dV}{dr}$ to help sketch the E-curve.

   E is simply the negative of the potential gradient.
2. The potential gradient near q_3 is steeper and thus the magnitude of E at q_3 should be larger than that at q_1.